API Security Risk Analysis Report

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Task: API Security Risk Analysis

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1. Introduction

APIs (Application Programming Interfaces) are widely used in modern applications such as mobile apps and web services. However, insecure APIs can expose sensitive data and lead to security risks. This report analyzes a public API to identify potential security issues.

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2. API Details

- **API Name:** JSONPlaceholder
- **Endpoint Tested:** https://jsonplaceholder.typicode.com/posts
- **Method Used:** GET

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3. API Testing

The API was tested using both a web browser and Postman.

The browser test confirmed that the API endpoint is publicly accessible without authentication.

The Postman test confirmed that the API responds successfully with status **200 OK** and returns structured JSON data.

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4. Observations

- The API returns data without requiring login or authentication

- The response contains user-related data (userId, title, body)
- No access control is implemented
- The API is openly accessible to anyone

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5. Identified Security Risks

- **Unauthenticated Access:** Anyone can access the API
- **Data Exposure:** API returns data without restrictions
- **No Authorization:** No user-level access control
- **No Rate Limiting:** API can be abused by sending multiple requests

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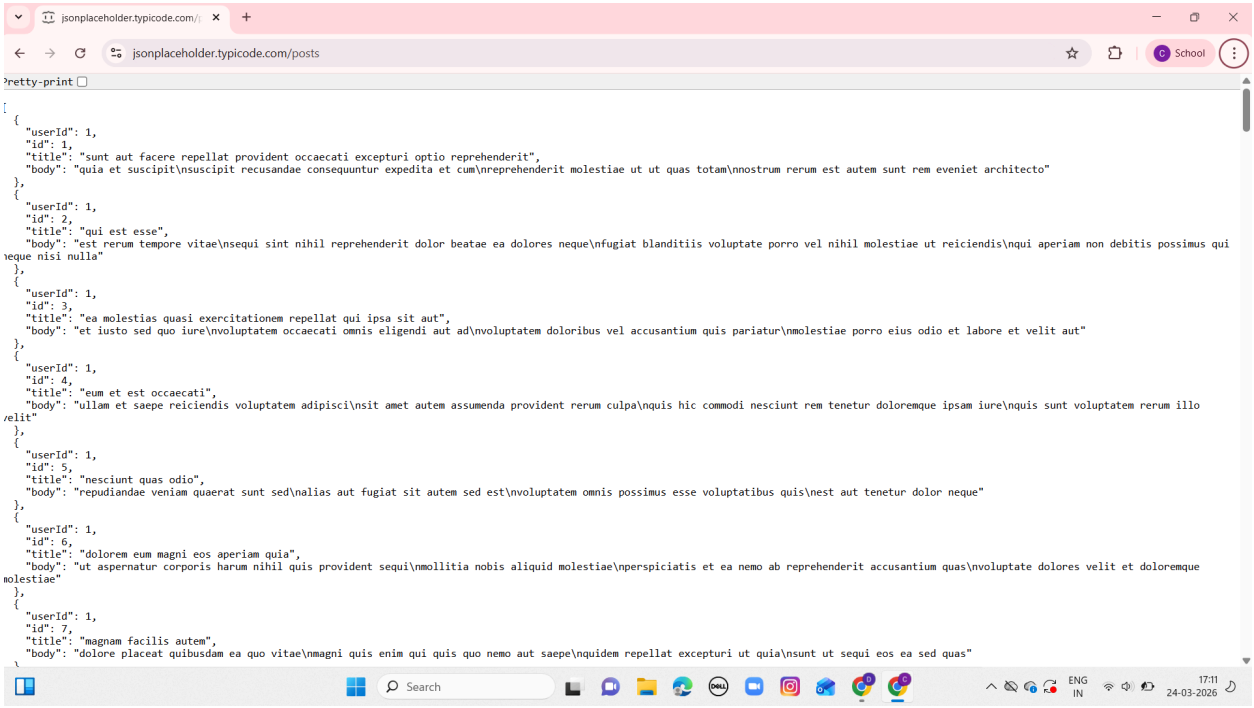
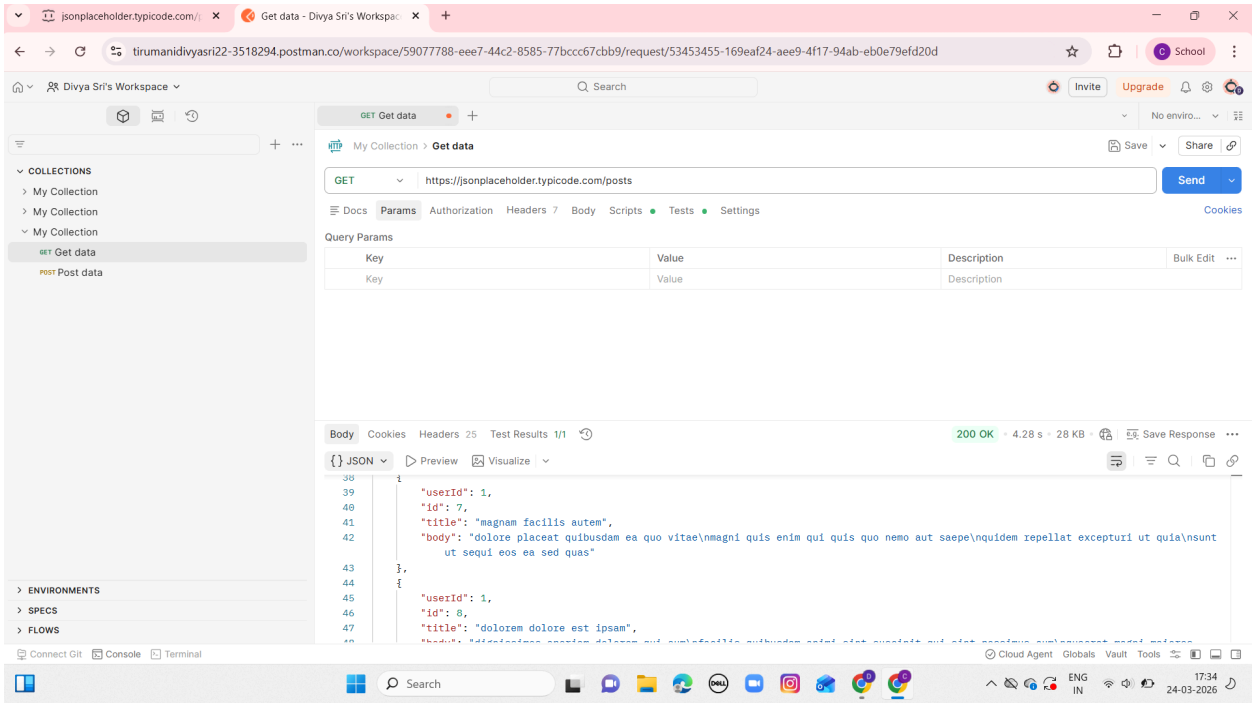
6. Evidence & Screenshots

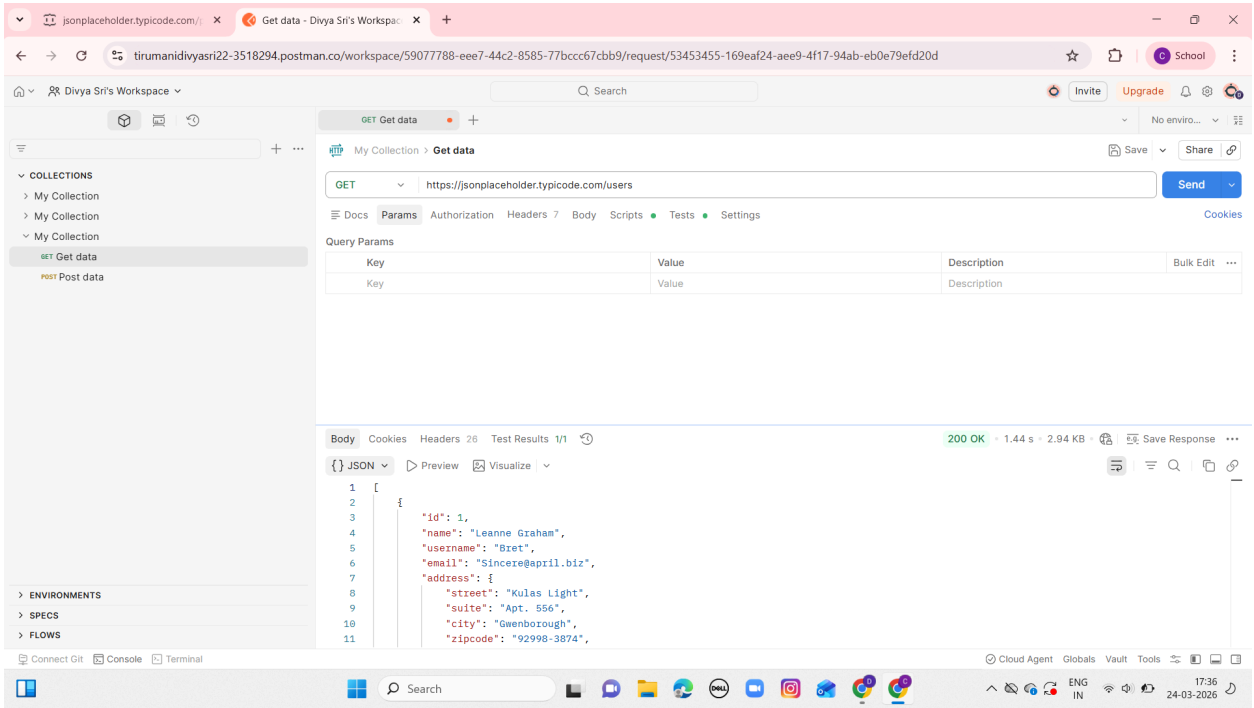
Screenshot 1: API response in browser



Screenshot 2: API response in Postman



Screenshot 3:



7. Risk Classification

Risk Level: MEDIUM

This API is publicly accessible and exposes data without authentication, which can lead to misuse if sensitive data is present.

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8. Prevention & Recommendations

- Implement authentication (API keys or tokens)
- Apply proper authorization controls
- Limit data exposure in API responses
- Enable rate limiting to prevent abuse
- Monitor API usage for suspicious activity

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9. Conclusion

The analysis shows that the API lacks authentication and access control. Although it is a demo API, similar issues in real systems can lead to serious security risks. Proper security measures must be implemented to protect API endpoints.